

Clmg: A Powerful C++ Library for Image Processing

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Clmg is a simple C++ toolkit intended to simplify the efforts of developers trying to implement new image-processing algorithms from scratch. A set of useful classes and functions that helps users to load/save, process and display various types of images is provided under a single header file `Clmg.h`.

Computer scientists or student community working on computer vision and image processing will find this tool extremely useful. Even if you are a programmer, occasionally coming across some level of image processing, this tool will be of great help as there is no standard C++ library for this purpose yet. The package is distributed under CeCILL licence.

Motivation behind Clmg

People who use image processing come from various scientific disciplines; we may find a doctor trying to have a better understanding of a patient's health condition with the diagnostic image of the patient's heart or a civil engineer trying to get an idea of a telemetric image of a dam to find out how strong it is. There could be a mathematician, biologist or computer programmer making use of image processing. A doctor might have a working knowledge of programming but we cannot expect him to know every minute detail of C++. Also, we cannot expect all users to deal with the same kind of images for their image-processing requirements.

So, the design of a general-purpose image-processing tool should cater to the needs of people work-

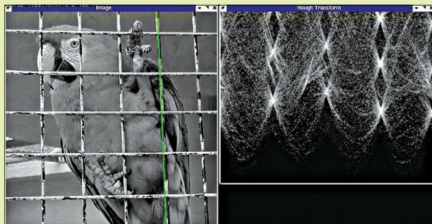


Fig. 1: Using Clmg for computation of Hough transform to detect lines in 2D images

Clmg classes

The namespace `Clmg` library encompasses all classes and methods. There are four classes included in `Clmg` library, in which only the first two have one template parameter.

Clmg. This class represents an image having width, height, depth and spectrum as dimensions. A simple linear buffer stores the pixel data.

ClmgList. This class represents a collection of images of varying sizes.

ClmgDisplay. This class represents a display window. We can have multiple windows and user interactions that could be managed through class methods.

ClmgException. When errors are encountered in the library method, `ClmgException` class throws exceptions.

ing on different kinds of images and having different programming capabilities. These designs should follow some basic rules that ensure simplicity, genericity, usefulness, extensibility, portability and freedom of use of the library. `Clmg` library was developed keeping in mind all these design considerations.

The framework

A handful of functions are included in order to write complex algorithms using minimal code. Let us take a look at some of these:

Supported file formats. `Clmg` supports a large number of input and output file formats. These include joint photographic experts group

(JPEG), 16-bit portable network graphics (PNG), 3D object file format, tagged image file format (TIFF) and more.

Image drawing. In order to facilitate drawing on images, there are many methods such as lines, polygons, ellipses, texts and vector fields.

Usual image-processing operators. These include filters, interpolators, fast Fourier transforms (FFTs), histograms, colour-base conversions and non-linear blur/sharpening, among others.

Arithmetic operators. Usual arithmetic operators like cosine, square-root and inverse tangent are available.

Expression evaluator. Synthetic